

# Real inflection points of perturbations of concurrent lines

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## Abstract

Let  $P$  be a real binary form of degree  $n \geq 3$  with  $n$  distinct real projective roots, and let  $V$  be a real homogeneous form of degree  $n$ , nonzero at the concurrent point. We give an exhaustive criterion for the real inflection points of  $P + \mu V = 0$ . If every real projective zero of the second polar  $V_{zz}$  on each limiting line is simple, then every sufficiently small nonzero real parameter gives a complex nonsingular curve whose ordinary real inflection points are in bijection with those zeros. A uniform estimate excludes real inflection points approaching the concurrent point. For every fixed  $P$ , explicit choices of the polar realize all even totals from 0 to  $n(n-2)$  in even degree and all totals  $n, n+2, \dots, n(n-2)$  in odd degree, within one rigid-isotopy class. In degree four we characterize all fourteen attainable cyclically labelled line-count vectors, and in even degree we determine the count on each oval. Radial profiles and the harmonic Stern family are applications. Two explicit quartics with the same second polar have six and eight ordinary real flexes, illustrating the role of simplicity. The complex specialization cycle is identified as an application of known divisor-specialization theory; the focus is the exhaustive real criterion and its explicit constructive use.

**Keywords:** Real algebraic curves; inflection points; Hessian; deformations; rigid isotopy.

## 1 Introduction and main results

A real inflection point, or flex, of a nonsingular real projective plane curve is a real point at which the tangent has intersection multiplicity at least three. The flex is *ordinary* when this multiplicity is exactly three. We count distinct real projective points. Nonsingularity below means nonsingularity over  $\mathbb{C}$ .

We study a concrete design problem: how can one determine and vary the real flex count of a perturbation of concurrent lines while keeping its topology under control? The maximal number  $n(n-2)$  allowed by Klein's bound and constructions attaining it are classical; see (Brugallé and López de Medrano, 2012, Theorem 1.1). Moreover, Brugallé–López de Medrano already construct odd-degree maximally inflected curves with all their real flexes on a pseudoline (Brugallé and López de Medrano, 2012, Proposition 7.2). Our question concerns a fixed line configuration and a specified perturbation, including counts below the maximum.

Fix  $n \geq 3$ , coordinates  $[x : y : z]$  on  $\mathbb{CP}^2$ , and  $O = [0 : 0 : 1]$ . Let

$$P = \prod_{i=1}^n \ell_i \in \mathbb{R}[x, y]_n, \quad F_\mu = P + \mu V, \quad C_\mu = \{F_\mu = 0\}, \quad (1)$$

where the real linear factors  $\ell_i$  are pairwise nonproportional and  $V \in \mathbb{R}[x, y, z]_n$ . Put

$$c = V(O), \quad S = V_{zz}, \quad L_i = \{\ell_i = 0\}. \quad (2)$$

The derivative  $V_{zz}$  is the second polar with respect to  $O$  in these coordinates, up to a nonzero conventional constant. It is homogeneous of degree  $n-2$ .

**Theorem 1.1** (Real polar criterion). *Assume  $c \neq 0$  and that every real projective zero of  $S|_{L_i}$  is simple, for each  $i$ . Write*

$$r_i = \#\{a \in L_i(\mathbb{R}) : S(a) = 0\}.$$

*For this fixed pair  $P, V$ , there is  $\varepsilon(P, V) > 0$  such that, for every real  $\mu$  with  $0 < |\mu| < \varepsilon(P, V)$ , the curve  $C_\mu$  is nonsingular and has exactly*

$$N_{\mathbb{R}}(C_\mu) = \sum_{i=1}^n r_i \tag{3}$$

*real flexes, all ordinary. They are the unique continuations of the real points of  $P = S = 0$ . No real flexes approach  $O$ .*

The simplicity assumption concerns only real zeros; multiple nonreal zeros are permitted. Since  $S(O) = n(n-1)c \neq 0$ , none of the restrictions vanishes identically and none has a zero at  $O$ . The statement includes points tending to  $z = 0$ , without requiring the flexes themselves to remain at infinity.

**Theorem 1.2** (Explicit attainable totals). *For every fixed  $P$  as above, the set of totals attainable under the hypotheses of Theorem 1.1 is*

$$\begin{cases} \{0, 2, 4, \dots, n(n-2)\}, & n \text{ even,} \\ \{n, n+2, n+4, \dots, n(n-2)\}, & n \text{ odd.} \end{cases} \tag{4}$$

*Explicit perturbations realizing all these totals can be chosen with  $V(O) = 1/[n(n-1)]$ . For a sufficiently small common positive parameter, the resulting curves belong to one rigid-isotopy class. In odd degree they have one pseudoline and no ovals; in even degree they have  $n/2$  nonnested ovals.*

Rigid isotopy here is a path of real degree- $n$  projective forms whose complex zero curves are nonsingular. Flexes may become nonordinary along such a path. Theorem 1.2 describes the transverse small-perturbation regime of (1); in particular its odd-degree lower endpoint does not assert a minimum for the entire rigid-isotopy class. The construction realizes all totals, not arbitrary independent choices of the vector  $(r_1, \dots, r_n)$ . Proposition 3.1 classifies these vectors in degree four, and Corollary 4.2 assigns the flexes to the individual ovals in even degree.

The global existence of all admissible real flex totals is classical: Ronga's Theorem II gives every even total from zero and every odd total from three up to the Klein bound (Ronga, 1998, Theorem II, p. 197). Our construction imposes additional constraints: the concurrent line configuration is prescribed, the perturbations are given explicitly, and the resulting curves can be chosen in one rigid-isotopy class. Thus the odd lower endpoint in our family describes its transverse small-parameter regime, not the global spectrum.

Changes of two real flexes through a contact of order four in generic families of nonsingular real curves are classical; see (Arroyo et al., 2014, Section 3.1). Our purpose is to determine the entire small-parameter real flex locus for a specified perturbation of a fixed concurrent line configuration, and to construct explicit profiles realizing the totals in (4).

Several ingredients have explicit antecedents. The strict sign of the binary Hessian is classical (Causa and Re, 2011). Second derivatives of the deformation occur in the determinant in (Esteves, 2010, Example 5.3), although that example does not directly apply to our concurrent configuration. Here the leading residual Hessian is computed directly as  $B_P V_{zz}$ . The complex cycle in Appendix A follows from (Esteves, 2010, Theorem 4.1) or

(Esteves et al., 2021, Proposition 5.2). The implicit function theorem, coefficient integration and compactness complete the argument once the uniform real exclusion estimate is proved.

Local theories of inflections under deformation provide complementary perspectives. Garay develops equivalence and versality with respect to inflections (Garay, 2002, Sections 3.2–3.6). Bruce, Fernandes and Tari study local inflection invariants; their finiteness criterion excludes germs containing a line (Bruce et al., 2025, Theorem 3.3(7) in the arXiv version). Our limiting curve is a union of straight lines. We divide the family Hessian by the parameter and use its residual equation, together with a uniform real estimate at the concurrent point. The contribution is the exhaustive real criterion for fixed  $P, V$  and its explicit constructive use.

## 2 The real polar criterion

### 2.1 A nonsingular rescaled model

**Lemma 2.1.** *If  $P$  has distinct projective roots and  $V(O) = c \neq 0$ , then  $C_\mu$  is nonsingular for every sufficiently small nonzero complex  $\mu$ . No polar-simplicity assumption is needed.*

*Proof.* Write

$$V = \sum_{j=0}^n V_j(x, y) z^{n-j}, \quad V_0 = c,$$

with  $V_j$  binary homogeneous of degree  $j$ . For  $\mu = u^n \neq 0$ , make the projective coordinate change  $[x : y : z] = [uX : uY : Z]$  and divide the equation by  $u^n$ . The transformed form is

$$G_u = P(X, Y) + \sum_{j=0}^n u^j V_j(X, Y) Z^{n-j}, \quad G_0 = P(X, Y) + cZ^n.$$

A singular point of  $G_0 = 0$  would have  $Z = 0$  and  $P_X = P_Y = 0$ . Euler's identity would then give a multiple projective root of  $P$ , a contradiction. The nonsingular locus in the coefficient space is open, so  $G_u = 0$  stays nonsingular for  $|u|$  small. Every small nonzero complex  $\mu$  has such an  $n$ -th root.

For real  $\mu = \sigma u^n$ , with  $u > 0$  and  $\sigma \in \{1, -1\}$ , the same real coordinate change gives instead

$$G_{\sigma, u} = P(X, Y) + \sigma \sum_{j=0}^n u^j V_j(X, Y) Z^{n-j}, \quad G_{\sigma, 0} = P(X, Y) + \sigma c Z^n. \quad (5)$$

Both real signs therefore have nonsingular real limiting models.  $\square$

The model  $P + cZ^n = 0$  also appears in the complex inflection count of Breiding et al. (2024, proof of Theorem 6.8, author-hosted version). For a prescribed perturbation  $V$ , the exhaustive real count below additionally uses the uniform exclusion estimate at the concurrent point.

### 2.2 The Hessian calculation

Set  $B_P = P_{xx}P_{yy} - P_{xy}^2$ . The ternary Hessian  $\mathcal{H}_\mu = \det \text{Hess}_{x,y,z} F_\mu$  satisfies the exact identity

$$\mathcal{H}_\mu = \mu B_P S + \mu^2 M + \mu^3 \det \text{Hess } V, \quad (6)$$

where

$$\begin{aligned} M = & (P_{xx}V_{yy} + P_{yy}V_{xx} - 2P_{xy}V_{xy})V_{zz} \\ & - P_{xx}V_{yz}^2 - P_{yy}V_{xz}^2 + 2P_{xy}V_{xz}V_{yz}. \end{aligned} \quad (7)$$

Indeed, the  $z$ -row and  $z$ -column of  $\text{Hess } P$  vanish; expanding the determinant by powers of  $\mu$  gives (6)–(7). Thus

$$K_\mu = \mathcal{H}_\mu / \mu = B_P S + \mu M + \mu^2 \det \text{Hess } V \quad (8)$$

extends polynomially to  $\mu = 0$ , with  $K_0 = B_P S$ . For  $\mu \neq 0$ , the zeros of  $K_\mu$  on  $C_\mu$  are its flexes.

For completeness, the local relation to contact multiplicity is especially simple. In an affine chart, let the curve be  $y = h(x)$  with  $F_y \neq 0$ , where  $F$  denotes the dehomogenized degree- $n$  form. Euler's identities and implicit differentiation give, on the curve,

$$\det \text{Hess}(F^{\text{hom}}) = (n-1)^2 F_y^3 h''. \quad (9)$$

Here the Hessian is dehomogenized in the same chart. The factor multiplying  $h''$  is a unit. A tangent contact of order  $k$  therefore contributes Hessian intersection multiplicity  $k-2$ , and a transverse Hessian intersection is an ordinary flex.

### 2.3 Uniform exclusion at the concurrent point

We include a proof of the classical binary-Hessian sign to keep the normalization and strictness explicit; compare (Causa and Re, 2011, Proposition 1 and proof of Theorem 1).

**Lemma 2.2.** *For the completely real squarefree binary form  $P$ ,  $B_P(v) < 0$  for every  $v \in \mathbb{R}^2 \setminus \{0\}$ . In particular,  $B_P \leq -\eta$  on the unit circle for some  $\eta > 0$ .*

*Proof.* Choose real linear coordinates putting the point in the chart  $x = 1$  and all roots in that chart, and write  $f(t) = P(1, t) = a \prod_i (t - a_i)$  with distinct real  $a_i$ . Away from the roots,

$$\begin{aligned} B_P(1, t) &= (n-1)(n f f'' - (n-1)(f')^2) \\ &= -(n-1)f(t)^2 \sum_{i < j} \left( \frac{1}{t - a_i} - \frac{1}{t - a_j} \right)^2 < 0. \end{aligned} \quad (10)$$

At a root the first expression equals  $-(n-1)^2 f'(t)^2 < 0$ . A real invertible linear coordinate change multiplies the Hessian determinant by the positive square of its determinant. Rescaling a nonzero vector also preserves the sign, since  $B_P$  has even degree  $2n-4$ . This proves the assertion in every direction; compactness gives  $\eta$ .  $\square$

Work near  $O$  in  $z = 1$ , and write  $(x, y) = \rho v$  with  $\rho > 0$  and  $v \in S^1$ . Since  $V(\rho v, 1)$  tends uniformly to  $c \neq 0$ , points of the real curve near  $O$  satisfy

$$|\mu| = \left| \frac{P(\rho v)}{V(\rho v, 1)} \right| \leq C \rho^n. \quad (11)$$

Uniformly in  $v$ , we have

$$B_P(\rho v) = \rho^{2n-4} B_P(v), \quad S(\rho v, 1) = n(n-1)c + O(\rho), \quad M(\rho v, 1) = O(\rho^{n-2}).$$

The second derivatives of  $V$  are bounded in this neighborhood. Combining these estimates with (8) and (11), on the curve, gives

$$\frac{K_\mu(\rho v, 1)}{\rho^{2n-4}} = n(n-1)c B_P(v) + O(\rho) + O(\rho^2) + O(\rho^4). \quad (12)$$

The leading term is uniformly separated from zero by Lemma 2.2. A fixed sufficiently small real neighborhood of  $O$  therefore contains no flexes of any small nonzero-parameter curve. Also  $O \notin C_\mu$ , since  $F_\mu(O) = \mu c$ . The estimate is uniform over all approach directions and uses no assumption on the relative rates at which  $\mu$  and  $\rho$  vanish.

## 2.4 Continuation and exhaustion

Let  $a$  be a real point of  $P = S = 0$ . It is not  $O$  and belongs to precisely one  $L_i$ . The differential of  $P$  is nonzero there. Moreover  $B_P(a) \neq 0$ , and the real simplicity of  $S|_{L_i}$  makes  $P = K_0 = 0$  a transverse intersection. The real analytic implicit function theorem applied in a projective chart to  $F_\mu = K_\mu = 0$  gives exactly one nearby real solution for every sufficiently small  $\mu$ . Its transverse intersection persists; (9) makes the flex ordinary. This also treats points on  $z = 0$ .

To exclude other real flexes, suppose that such points occur for a sequence of nonzero real parameters tending to zero. Compactness of  $\mathbb{RP}^2$  gives a convergent subsequence. Its limit lies on  $P = 0$ , and (12) excludes  $O$ . At any other real point,  $K_0 = B_P S$  and  $B_P \neq 0$ , so the limit must be a real zero of  $S|_{L_i}$ . The local uniqueness just proved is a contradiction. Together with Lemma 2.1, this proves Theorem 1.1.

## 3 Constructing all attainable totals

Write  $d = n - 2 = 2a + \epsilon$  with  $\epsilon \in \{0, 1\}$ . For each  $i$ , the restriction  $S|_{L_i}$  has projective degree  $d$ . Nonreal roots come in conjugate pairs, counted with multiplicity, whereas its real roots are simple. Consequently

$$r_i \equiv d \pmod{2}, \quad \epsilon \leq r_i \leq d.$$

Theorem 1.1 gives the containment in (4). We now realize every total in that set for the specified  $P$ .

Choose unit vectors  $v_i$  on the  $n$  distinct line directions. There is a real linear form  $\lambda$  such that

$$\lambda_i = \lambda(v_i)^2 > 0 \quad \text{are pairwise distinct.}$$

Indeed, it suffices to avoid the finitely many proper linear conditions  $\lambda(v_i) = 0$ ,  $\lambda(v_i - v_j) = 0$ , and  $\lambda(v_i + v_j) = 0$  in the two-dimensional dual space. The latter vectors are nonzero because the directions are distinct.

Choose pairwise distinct real numbers  $t_1, \dots, t_a$ , none equal to a  $\lambda_i$ , and define

$$S = z^\epsilon \prod_{j=1}^a (z^2 + t_j(x^2 + y^2) - \lambda(x, y)^2). \quad (13)$$

On  $L_i$ , with coordinates  $[x : y : z] = [\rho v_i : z]$ , each factor is  $z^2 + (t_j - \lambda_i)\rho^2$ . It gives two simple real roots when  $t_j < \lambda_i$ , and none when  $t_j > \lambda_i$ . The factor  $z^\epsilon$  adds  $\epsilon$  simple roots. Roots belonging to different factors cannot coincide because the  $t_j$  are distinct. Also  $S(O) = 1$ .

For one  $t_j$ , the number  $b_j = \#\{i : \lambda_i > t_j\}$  can be any integer from 0 to  $n$ . Thus

$$\sum_i r_i = \epsilon n + 2 \sum_{j=1}^a b_j. \quad (14)$$

Every  $K \in \{0, \dots, an\}$  is a sum of  $a$  integers in  $\{0, \dots, n\}$ : write  $K = qn + r$ , use  $q$  copies of  $n$ , the nonzero remainder if present, and zeros. When  $q = a$ , the remainder is zero. Repeated values of  $b_j$  are realized by distinct  $t_j$  in the same open interval. Formula (14) therefore gives every desired total. For  $n = 3$  the product is empty,  $S = z$ , and the same argument gives the sole total 3.

There is no extension or interpolation problem for these restrictions: we have constructed a single homogeneous  $S$ . If  $S = \sum_{j=0}^{n-2} s_j(x, y)z^{n-2-j}$ , set

$$V = \sum_{j=0}^{n-2} \frac{s_j(x, y)z^{n-j}}{(n-j)(n-j-1)}. \quad (15)$$

Then  $V_{zz} = S$  and  $V(O) = 1/[n(n-1)]$ . Terms  $A_{n-1}(x, y)z + B_n(x, y)$  may be added without changing the limiting criterion. The topology and common-parameter assertions of Theorem 1.2 follow from the next section.

| Degree | Radial totals | Totals from (13)    |
|--------|---------------|---------------------|
| 4      | 0, 8          | 0, 2, 4, 6, 8       |
| 5      | 5, 15         | 5, 7, 9, 11, 13, 15 |
| 6      | 0, 12, 24     | 0, 2, 4, ..., 24    |

Table 1: Attainable totals in the respective transverse small-parameter regimes for each fixed line configuration. The radial column follows from Corollary 5.1.

**Proposition 3.1** (Quartic line-count vectors). *Fix four distinct concurrent real lines, labelled in cyclic order in  $\mathbb{RP}^1$ . Among perturbations satisfying Theorem 1.1, the attainable vectors  $(r_1, r_2, r_3, r_4)$  are exactly the fourteen members of  $\{0, 2\}^4$  other than*

$$(2, 0, 2, 0), \quad (0, 2, 0, 2).$$

*Each can be realized with  $V(O) = 1/12$ . The labels are retained: no quotient by cyclic rotation or permutation is taken.*

*Proof.* Write a general second polar as

$$S = az^2 + zL(x, y) + Q(x, y), \quad \Delta_S = L^2 - 4aQ, \quad a = 12V(O) \neq 0,$$

where  $L, Q$  have degrees one and two. For a representative  $v_i$  of the direction of  $L_i$ , the restricted form in coordinates  $[\rho : z]$  is  $az^2 + \rho L(v_i)z + \rho^2 Q(v_i)$ . It is nonzero at  $\rho = 0$ . Simplicity of its real zeros is equivalent to  $\Delta_S(v_i) \neq 0$ , and  $r_i = 2$  or  $0$  according as this value is positive or negative.

A nonzero binary quadratic has at most two distinct zeros in  $\mathbb{RP}^1$ . Thus its signs at four cyclic directions cannot alternate, since alternation would require a zero in each of four intervening intervals. Conversely, every nonalternating sign pattern has zero or two cyclic sign changes. A definite quadratic realizes either constant pattern. For two sign changes, choose one zero direction in each of the two corresponding open intervals and take the product of linear forms vanishing there, with the appropriate overall sign. This realizes the pattern with nonzero values at all four given directions.

For any such quadratic  $\Delta_S$ , set

$$S = z^2 - \frac{\Delta_S}{4}, \quad V = \frac{z^4}{12} - \frac{\Delta_S z^2}{8}.$$

Then  $V_{zz} = S$ ,  $V(O) = 1/12$ , and the discriminant of the restriction is precisely  $\Delta_S(v_i)$ . Theorem 1.1 proves sufficiency for every fixed  $P$ .  $\square$

## 4 Topology and rigid isotopy

**Proposition 4.1.** *Fix  $P$ ,  $c \neq 0$ , and  $\sigma \in \{1, -1\}$ . For every fixed  $V$  with  $V(O) = c$ , the curves  $P + \mu V = 0$ , for sufficiently small real  $\mu$  of sign  $\sigma$ , are rigidly isotopic to*

$$P(X, Y) + \sigma c Z^n = 0. \tag{16}$$

*In odd degree their real locus is one pseudoline; in even degree  $n = 2m$  it consists of  $m$  nonnested ovals. For a compact segment of such profiles, a common small parameter gives a path of nonsingular curves. Polar simplicity need not hold along this path.*

*Proof.* The rescaled path (5) consists of nonsingular forms down to  $u = 0$ , after shrinking the parameter interval. For  $u > 0$ , the real coordinate transformation  $[X : Y : Z] \mapsto [uX : uY : Z]$  is connected to the identity through invertible real transformations. This proves rigid isotopy to (16).

The model misses  $O$ , so projection to  $[X : Y] \in \mathbb{RP}^1$  is defined. For odd  $n$ , each direction has exactly one real value of  $Z$ , up to projective scaling. The projection is a continuous bijection from a compact space to  $\mathbb{RP}^1$ , hence a homeomorphism. A nonsingular odd-degree curve has nontrivial mod-two projective homology. Its sole real component is therefore a pseudoline.

For even  $n = 2m$ , the sign of  $P$  is well-defined on directions after choosing unit representatives. The function  $-P/(\sigma c)$  is positive on  $m$  alternate intervals bounded by consecutive roots in  $\mathbb{RP}^1$ . On one such closed interval  $I$ , choose a continuous unit representative  $v(\theta)$  and put

$$f(\theta) = \left( -\frac{P(v(\theta))}{\sigma c} \right)^{1/n}.$$

The branches  $Z = \pm f(\theta)$  meet at the endpoints and bound

$$D_I = \{[v(\theta) : Z] : \theta \in I, |Z| \leq f(\theta)\}. \quad (17)$$

The strip with its two endpoint fibers collapsed is a closed disk. Its projective map is injective: distinct interior directions remain distinct, and a fixed unit representative identifies no two values of  $Z$ . Thus  $D_I$  is an embedded disk, missing  $O$ . The intervals  $I$  are disjoint, including their endpoints, and so are the disks. Their boundaries give precisely  $m$  nonnested ovals.

Finally, let  $V_s$ ,  $s \in [0, 1]$ , be a segment with  $V_s(O) = c$ . The rescaled forms converge in coefficient space to (16) uniformly in  $s$ . Openness of the nonsingular locus gives a common bound on  $u$ . Hence one small nonzero real  $\mu$  of sign  $\sigma$  makes the entire segment nonsingular. No simplicity of  $V_{s,zz}$  is used for this step.  $\square$

Choose one profile from Section 3 for each desired total, and join them by finitely many profile segments. Taking the minimum of their smoothness thresholds and the endpoint flex thresholds gives a single positive  $\mu$  realizing all totals and all connecting paths. This completes Theorem 1.2. The same finite-choice argument places representatives of all fourteen vectors in Proposition 3.1 in one rigid-isotopy class at a common sufficiently small positive parameter.

In odd degree the transformation  $[X : Y : Z] \mapsto [X : Y : -Z]$  also connects the two model signs. It belongs to the identity component of  $\mathrm{PGL}_3(\mathbb{R})$ : multiply its matrix by  $-1$  and rotate the  $X, Y$ -plane through  $\pi$ . Positive rescaling of  $Z$  adjusts the magnitude of  $c$ . For fixed  $P$ , all sufficiently small odd-degree profiles with  $V(O) \neq 0$ , of either parameter sign, consequently lie in the same rigid-isotopy class. Their antipodal lifts to  $S^2$  are single embedded circles with two complementary domains. This topological statement does not assert that nonharmonic profiles are spherical harmonics.

**Corollary 4.2** (Flexes on each oval). *Assume the hypotheses of Theorem 1.1, with  $n$  even, and fix the parameter sign  $\sigma$ . For sufficiently small  $\mu$  of this sign, label each oval by the interval  $I_{ij}$  between consecutive directions  $i, j$  on which  $-P/(\sigma c) > 0$ . The oval has exactly*

$$N_{ij} = r_i + r_j$$

*real flexes. In particular  $N_{ij}$  is even and at most  $2(n-2)$ . If the total is  $n(n-2)$ , every oval has  $2(n-2)$  real flexes.*

*Proof.* Write  $\mu = \sigma u^n$ ,  $u > 0$ . The rescaled curves (5) converge smoothly to (16), whose ovals have disjoint tubular neighborhoods. The implicit function theorem and compactness identify, for all small  $u$ , the perturbed real locus in each neighborhood with a unique normal graph over its model oval, with no additional real components.

Let a continued flex tend to  $a \in L_i(\mathbb{R}) \setminus \{O\}$ . Choose bounded local homogeneous representatives  $[x(\mu) : y(\mu) : z(\mu)]$  with a nonzero limiting  $(x, y)$ -pair in direction  $v_i$ . In the rescaled coordinates,

$$[x(\sigma u^n) : y(\sigma u^n) : uz(\sigma u^n)] \longrightarrow [v_i : 0].$$

This endpoint belongs to precisely one model oval, namely that of the unique admissible interval adjacent to direction  $i$ . Consequently all  $r_i$  continued flexes lie on that oval, including those whose unscaled limit is at infinity. Its other endpoint contributes  $r_j$ . Exhaustion in Theorem 1.1 gives the formula. Finally,  $0 \leq r_i \leq n - 2$  and  $r_i$  is even; at the maximal total every  $r_i$  equals  $n - 2$ .  $\square$

For  $n = 4$ , maximal curves in this transverse family therefore have inflection type  $(4, 4)$ . The global classification of maximally inflected quartics also includes  $(6, 2)$  and  $(8, 0)$  (Brugallé and López de Medrano, 2012, Proposition 7.1); the marked line vectors in Proposition 3.1 classify a different object.

Reversing the parameter sign changes the pairing of consecutive directions. It preserves the total but can change the counts assigned to the resulting ovals. No constancy of these counts along arbitrary rigid isotopies is asserted.

## 5 Radial profiles and the Stern family

### 5.1 The radial criterion in both parities

Let  $m = \lfloor n/2 \rfloor$ , and write

$$Q(q) = \sum_{k=0}^m c_k q^k, \quad V_Q = \sum_{k=0}^m c_k (x^2 + y^2)^k z^{n-2k}. \quad (18)$$

Its second polar is encoded in the affine chart  $z = 1$  by

$$\begin{aligned} D_n(q) &= 4q^2 Q'' + (6 - 4n)qQ' + n(n-1)Q \\ &= \sum_{k=0}^{m-1} (n-2k)(n-2k-1)c_k q^k. \end{aligned} \quad (19)$$

In particular  $S(x, y, 1) = D_n(x^2 + y^2)$ . The coefficient  $c_m$  disappears in both parities.

**Corollary 5.1.** *Assume  $c_0 c_{m-1} \neq 0$  and every positive root of  $D_n$  is simple. If  $r$  is the number of positive roots, then for every sufficiently small nonzero real  $\mu$ , the curve  $P + \mu V_Q = 0$  is nonsingular and has exactly*

$$\begin{cases} n(2r+1), & n = 2m+1, \\ 2nr, & n = 2m \end{cases}$$

*ordinary real flexes. The finite limiting points have squared radii equal to the positive roots of  $D_n$ . In odd degree there is also one flex tending to infinity on each line; in even degree there are no such limiting flexes.*

*Proof.* On each line, the affine polar has two simple roots for each positive root of  $D_n$ . In odd degree its homogenization has a simple factor  $z$  at infinity, with nonzero residual coefficient  $6c_{m-1}$ . In even degree the value at a real infinite direction is  $2c_{m-1}(x^2 + y^2)^{m-1} \neq 0$ . Theorem 1.1 applies.  $\square$

This gives all radial totals in Table 1. More generally, given  $D_n = \sum_{k=0}^{m-1} d_k q^k$  with  $d_0 d_{m-1} \neq 0$ , choose

$$c_k = \frac{d_k}{(n-2k)(n-2k-1)} \quad (k < m), \quad c_m \text{ arbitrary.} \quad (20)$$

Choosing  $r$  distinct positive and  $m-1-r$  distinct negative roots realizes any  $r = 0, \dots, m-1$ , and prescribes the limiting squared radii. Repeated negative or nonreal roots are allowed. Common roots of  $Q$  and  $D_n$  are also allowed when the positive roots of  $D_n$  remain simple.

For odd degree the real locus is a pseudoline by Proposition 4.1, now without any equal-spacing assumption. For even degree, Corollary 4.2 gives  $4r$  flexes on each oval, since every line contributes  $2r$ . The fixed- $P$  isotopy statement of Proposition 4.1 applies to both radial and nonradial profiles.

## 5.2 Equal spacing, exact locations and triple roots

For odd  $n = 2m + 1$ , put  $W_n = \text{Im}(x + iy)^n$ . For the affine equation  $W_n + \mu Q(x^2 + y^2) = 0$ , let

$$T(q) = 4\{2q(Q')^3 + n(n-3)Q(Q')^2 - n^2Q^2Q''\}, \quad L_\mu(q) = n^2q^{n-2}D_n(q) + \mu^2T(q).$$

At real curve points away from the origin, the tangent-Hessian numerator is

$$N = \mu L_\mu(q). \quad (21)$$

Here  $N = f_y^2 f_{xx} - 2f_x f_y f_{xy} + f_x^2 f_{yy}$ , with  $f = W_n + \mu Q(x^2 + y^2)$ , is the affine Hessian evaluated on the tangent vector  $(-f_y, f_x)$ . To check the formula, write  $W_n = \rho^n p(\theta)$ , use  $p'' = -n^2 p$ ,  $(p')^2 = n^2(1 - p^2)$ , and substitute  $\rho^n p = -\mu Q$  in the tangent quadratic form of the affine Hessian. For example, with  $A = 2qQ' - nQ$ , the cubic coefficient simplifies by

$$A^3 + n^2QA^2 - n^2Q^2D_n = q^2T.$$

The curve equation is feasible exactly when  $\Delta_\mu(q) = q^n - \mu^2Q(q)^2 \geq 0$ . Near a simple positive root  $\alpha$  of  $D_n$ , this inequality is strict for small  $\mu$ . Its  $2n$  flexes have the common squared radius

$$q_\alpha(\mu^2) = \alpha - \frac{T(\alpha)}{n^2\alpha^{n-2}D'_n(\alpha)}\mu^2 + O(\mu^4), \quad (22)$$

determined by  $L_\mu = 0$ . Their angles satisfy  $\sin(n\theta) = -\mu Q(q_\alpha)/q_\alpha^{n/2}$ . Local uniqueness accounts for all  $2n$  points.

The  $n$  remaining flexes in this odd sectoral family lie exactly at infinity. After rotating any limiting direction to  $[1 : 0 : 0]$ , the polynomial  $W_n(1, y)$  is odd in  $y$ . The radial perturbation is even in  $y$  and odd in  $z$ . The implicit graph  $y = h(z, \mu)$  is therefore odd in  $z$ , so  $h''(0, \mu) = 0$ ; ordinary contact follows from Corollary 5.1. Section 6.2 shows why these exact locations do not survive general line configurations.

There is a simple exact-location observation that does not require equal spacing. Suppose  $Q$  has a root of multiplicity exactly three at  $\alpha > 0$ . At each of the  $2n$  intersections of  $P = 0$  with  $x^2 + y^2 = \alpha$ ,  $Q = Q' = Q'' = 0$ . The curve is regular there because its transverse derivative equals that of  $P$ . Its tangent is the corresponding limiting line, whose restriction of the equation is  $\mu Q(x^2)$  in an orthonormal line coordinate. For  $\mu \neq 0$ , the contact order is exactly three. These points are fixed ordinary flexes, rather than hyperflexes. Consistently,  $D_n(\alpha) = 0$  and  $D'_n(\alpha) = 4\alpha^2Q'''(\alpha) \neq 0$ .

The following degree-seven profiles illustrate the permitted degeneracies and the distinction between roots of  $Q$  and  $D_7$ :

| $Q(q)$                                        | $D_7(q)$        | $N_{\mathbb{R}}$ |
|-----------------------------------------------|-----------------|------------------|
| $\frac{1}{42} + \frac{q}{10} + \frac{q^2}{6}$ | $(q+1)^2$       | 7                |
| $1 - 7q^2 + q^3$                              | $42(1 - q^2)$   | 21               |
| $(q-1)^3$                                     | $-6(q-1)(3q-7)$ | 35               |

The first has vanishing top radial coefficient and a double negative root of  $D_7$ . In the last,  $q = 1$  is a fixed circle of fourteen ordinary flexes for  $W_7$ ; the other positive root of  $D_7$  is  $7/3$ .

### 5.3 Harmonic specialization

Let  $L_n$  be the Legendre polynomial normalized by  $L_n(1) = 1$ , write  $R^2 = x^2 + y^2 + z^2$ , and put

$$Z_n = R^n L_n(z/R).$$

This expression is a homogeneous polynomial. For odd  $n = 2m + 1$ , the classical sectoral and zonal harmonics give the Stern family

$$W_n + \mu Z_n = 0; \tag{23}$$

see Bérard–Helffer ([Bérard and Helffer, 2016](#)). The lowering identity

$$(Z_n)_{zz} = n(n-1)Z_{n-2} \tag{24}$$

can be checked without importing a new geometric assertion. The generating function

$$\sum_{k \geq 0} Z_k t^k = (1 - 2zt + (x^2 + y^2 + z^2)t^2)^{-1/2}$$

satisfies  $\partial_z \sum Z_k t^k = t \sum Z_k t^k + t^2 \partial_t \sum Z_k t^k$ . Comparing coefficients gives  $\partial_z Z_k = k Z_{k-1}$ ; applying it twice proves (24).

In the chart  $z = 1$ ,

$$Q_n(q) = (1+q)^{n/2} L_n((1+q)^{-1/2}), \quad D_n(q) = n(n-1)Q_{n-2}(q).$$

Equivalently, the explicit radial coefficients are

$$Q_n(q) = \sum_{k=0}^m \frac{(-1)^k n!}{4^k (k!)^2 (n-2k)!} q^k.$$

The odd Legendre polynomial  $L_{n-2}$  has  $m-1$  positive simple roots  $\tau_j \in (0, 1)$ ; its remaining central root is zero. Hence  $Q_{n-2}$  has exactly  $m-1$  positive simple roots  $\tau_j^{-2} - 1$ , and its leading coefficient is nonzero. Also  $Q_n(0) = 1$ . Corollary 5.1 gives exactly  $n(n-2)$  ordinary real flexes for all sufficiently small nonzero real  $\mu$ . The finite limiting squared radii are  $\tau_j^{-2} - 1$ .

For clarity, the elementary Legendre root property used here follows from orthogonality on  $[-1, 1]$ : if a degree- $k$  Legendre polynomial had fewer than  $k$  sign-changing roots in  $(-1, 1)$ , multiplication by the product of their linear factors would give a polynomial of degree less than  $k$  with a nonzero integral against it, contradicting orthogonality. Thus all  $k$  roots are simple and internal; parity supplies the positive-root count.

All these real flexes lie on the sole pseudoline, and its spherical lift has two complementary domains. The nodal construction of the harmonic family is classical ([Bérard and Helffer, 2016](#), Proposition 1(i), preprint version); the present specialization concerns the exhaustive flex calculation. Maximal existence and concentration on a pseudoline are not asserted as new.

## 6 Examples and the limits of the criterion

### 6.1 A nonradial quartic with six real flexes

Take

$$P = y(x - y)(x - 2y)(x - 3y), \quad V = \frac{z^4}{12} + \frac{(-x^2 + 2y^2)z^2}{2}. \quad (25)$$

Then  $S = z^2 - x^2 + 2y^2$  has respectively 2, 0, 2, 2 real projective zeros on the four lines. Theorem 1.1 therefore gives six ordinary real flexes at sufficiently small nonzero parameter. This total is unavailable to the transverse degree-four radial profiles of Corollary 5.1, which give zero or eight.

The fixed value  $\mu = 1/1000$  is also certified directly. Multiplying  $F_\mu$  by 12000 gives

$$F = 12000y(x - y)(x - 2y)(x - 3y) + z^4 - 6x^2z^2 + 12y^2z^2.$$

Let  $f = F(x, y, 1)$  and let  $h$  be the dehomogenized primitive ternary Hessian. Elimination in  $y$  gives a squarefree polynomial  $E(x)$  of degree 24. The penultimate subresultant is linear in  $y$ , with coefficient coprime to  $E$ ; the leading coefficient of  $f$  in  $y$  is the nonzero constant  $-72000$ . Sturm's method gives exactly six real roots of  $E$ , hence six real affine intersection points. There are 24 distinct complex affine intersections, the full Bézout degree, so all are simple. Separate exact checks show that the curve is nonsingular and that none of its four points at infinity is a flex. The accompanying script `examples/check_quartic.py` reconstructs these polynomials and checks, using rational arithmetic. This fixed-parameter certificate is distinct from the small-parameter theorem.

### 6.2 An asymmetric quintic and moving locations

Let

$$P = y(x - y)(x - 2y)(x - 3y)(x - 4y).$$

Near  $y = 0$ , write  $P = dx^4y + ex^3y^2 + O(y^3)$ ; here  $d = 1$ ,  $e = -10$ . For  $Q = 1 - q$ , the polynomial  $D_5 = 20 - 6q$  has positive root  $\alpha = 10/3$ . The radial corollary gives fifteen flexes. The local graph  $y = \mu h_1(x) + \mu^2 h_2(x) + O(\mu^3)$  has

$$h_1 = -\frac{1 - x^2}{dx^4}, \quad h_2 = -\frac{e(1 - x^2)^2}{d^3x^9}.$$

At  $x_0 = \pm\sqrt{10/3}$ ,

$$h_1'''(x_0) = \frac{12}{d}x_0^{-5}, \quad h_2''(x_0) = -\frac{50e}{d^3}x_0^{-11}.$$

Solving  $h_{xx} = 0$  gives  $x_{\text{fl}} = x_0 + 25e x_0^{-6} \mu / (6d^2) + O(\mu^2)$ . Consequently the squared radii of the two flexes near this line are

$$q_+(\mu) = \frac{10}{3} - \frac{3\sqrt{30}}{4}\mu + O(\mu^2), \quad q_-(\mu) = \frac{10}{3} + \frac{3\sqrt{30}}{4}\mu + O(\mu^2). \quad (26)$$

They have a common limiting radius but distinct radii for small nonzero parameter.

For the same  $P$ , take  $Q = 1 + q + q^2$ . Now  $D_5 = 20 + 6q$ , so the five real flexes all tend to infinity. In the chart  $x = 1$  at  $[1 : 0 : 0]$ , write  $y = h(z, \mu)$ . Implicit differentiation gives

$$h_{zz}(0, \mu) = 20\mu^2, \quad z_{\text{fl}}(\mu) = \frac{10}{3}\mu + O(\mu^2).$$

This flex therefore leaves  $z = 0$ . Both effects follow by substitution and differentiation; no count is inferred from a numerical plot.

Intermediate quintic totals also have short formulas. For this  $P$ , the polars

$$S = z(z^2 - x^2 + \lambda y^2)$$

with  $\lambda = 2, 5, 10, 17$  have respectively 13, 11, 9, 7 simple real zeros on the line configuration. Integrating twice in  $z$  and applying Theorem 1.1 realizes these totals. These are statements at sufficiently small parameter, not additional fixed-parameter certificates.

### 6.3 The same polar with six or eight real flexes

The polar alone need not determine the real count when a restricted root is multiple. Consider

$$P = y(x - y)(x - 2y)(x - 3y), \quad V_b = \frac{z^4}{12} + \frac{(y^2 - x^2)z^2}{2} + bx^4. \quad (27)$$

For every  $b$ , we have  $V_b(O) = 1/12$  and

$$V_{b,zz} = z^2 - x^2 + y^2.$$

This polar has two simple real zeros on each of  $y = 0, x = 2y, x = 3y$ , and a double zero at  $a = [1 : 1 : 0]$  on  $x = y$ . We show that for sufficiently small  $\mu > 0$ , the choice  $b = 1$  gives six ordinary real flexes and  $b = -1$  gives eight.

In the chart  $x = 1$ , put  $p(y) = y(1 - y)(1 - 2y)(1 - 3y)$ . Since  $p'(1) = -2$ , the local curve is  $y = h(z, \mu)$ , where  $h$  is analytic and even in  $z$ . Write  $h = 1 + \mu h_1 + \mu^2 h_2 + O(\mu^3)$ . Substitution gives

$$\begin{aligned} h_1 &= \frac{b}{2} + \frac{z^4}{24}, \\ h_2 &= -\frac{9b^2}{8} - \frac{3bz^4}{16} + \frac{bz^2}{4} - \frac{z^8}{128} + \frac{z^6}{48}. \end{aligned}$$

The normalized flex equation is therefore

$$\frac{h_{zz}}{\mu} = \frac{z^2}{2} + \frac{b\mu}{2} + O(\mu z^2 + \mu^2). \quad (28)$$

The quotient extends analytically to  $\mu = 0$ . By parity it is analytic in  $w = z^2$ , with derivative in  $w$  close to  $1/2$ . For  $b = 1$  it is positive for real  $z$  in a fixed small neighborhood and small  $\mu > 0$ . For  $b = -1$  it has exactly one nearby zero in  $w$ , namely  $w = \mu + O(\mu^2) > 0$ , yielding two real zeros  $z = \pm\sqrt{\mu} + O(\mu^{3/2})$ . They are simple zeros of  $h_{zz}$ , hence ordinary flexes.

The six simple polar zeros continue as ordinary real flexes. The nonsingularity lemma and the central exclusion estimate do not require polar simplicity. Together with projective compactness and the above local calculation, they exclude all further real flexes for these fixed profiles.

At  $b = 0$ ,  $a$  is a fixed smooth point with tangent  $y = x$ . Restricting the equation to that tangent gives  $\mu z^4/12$ , so the contact order is exactly four. Locally  $h - 1$  is divisible by  $z^4$ , and  $h_{zz}/(\mu z^2)$  extends to a nonzero analytic function with value  $1/2$  at  $(0, 0)$ . Thus there are six ordinary real flexes and this one hyperflex: seven distinct points, with total real Hessian intersection multiplicity eight.

There are also exact certificates at  $\mu = 1/1000$ . For  $b = 1, -1$ , respectively, the integer forms

$$12000P + z^4 + 6(y^2 - x^2)z^2 + 12bx^4$$

have six and eight ordinary real flexes. The accompanying `examples/check_extensions.py` verifies complex nonsingularity, no flex at infinity, a squarefree elimination polynomial of

degree 24, coprimality of the linear subresultant coefficient, and the real-root count by Sturm's theorem. These conditions identify all 24 distinct complex curve–Hessian intersections and certify ordinarity. The exact output is `EXTENSION_CHECKS.json`.

The segment  $b \in [-1, 1]$  has fixed central value, so Proposition 4.1 joins these profiles in one rigid-isotopy class at a common sufficiently small parameter. The two certificates at  $1/1000$  do not certify the whole segment at that value. The change through an order-four contact is classical (Arroyo et al., 2014, Section 3.1); here the example keeps the second polar fixed and exhibits the effect of a coefficient lost under two differentiations.

## 6.4 Colliding directions and tropical comparison

The example above concerns a multiple restricted polar root. It is distinct from a triple root of  $Q$  in Section 5.2, where the relevant root of  $D_n$  may still be simple.

Uniformity holds on the compact profile segments in Proposition 4.1. It is not asserted for unbounded coefficient families or colliding line directions. If two directions merge, the rescaled model can become singular and the strict binary-Hessian bound can degenerate. The theorem therefore supplies no positive common threshold or assertion about persistence of distinct flexes through that limit.

For the natural radial degeneration  $\mu = t$ , after a generic rotation of  $x, y$ , the degree- $n$  edge coefficients have valuation zero and the nonzero lower radial coefficients have valuation one. The lower Newton hull has a coarse triangular cell of normalized area  $n^2$ . It is not the unimodular subdivision assumed in the nonsingular tropical lifting statements of (Brugallé and López de Medrano, 2012, Theorems 5.6–5.7). Those particular hypotheses do not apply directly to this presentation. This observation does not rule out other tropical presentations and is not evidence of novelty. Our real criterion and construction require no tropical lifting assertion.

## A The weighted complex specialization cycle

For  $\mu \neq 0$ , let  $\text{Infl}(C_\mu)$  be the intersection cycle of the nonsingular curve with its ternary Hessian, including multiplicities. With  $S = V_{zz}$  and  $c \neq 0$ ,

$$\lim_{\mu \rightarrow 0} \text{Infl}(C_\mu) = 2n(n-2)[O] + (P \cdot S). \quad (29)$$

Here  $P \cdot S$  denotes the projective intersection cycle, of degree  $n(n-2)$ , supported away from  $O$ . No ordinary-flex assumption on its nonreal points is made.

We give a direct specialization argument. The forms  $P$  and  $K_0 = B_P S$  have no common component:  $S(O) \neq 0$  excludes any  $L_i$  from  $S$ , and the restriction of  $B_P$  to a simple-root direction of  $P$  is nonzero. The family

$$\mathcal{Z} = \{F_\mu = K_\mu = 0\} \subset \mathbb{CP}^2 \times \Delta$$

over a sufficiently small complex disk has finite special fiber. Properness and upper semicontinuity of fiber dimension make the projection finite after restricting  $\Delta$ . At a point above  $\mu = 0$ , the three elements  $\mu, F_\mu, K_\mu$  form a system of parameters in the regular local ambient ring of dimension three. They are a regular sequence, also in the order  $F_\mu, K_\mu, \mu$ . Thus the finite family has no  $\mu$ -torsion and is flat over the one-dimensional smooth base. Its special intersection cycle is the specialization of the generic one.

On each  $L_i$ ,  $B_P$  is a nonzero constant times  $\rho^{2n-4}$ , where  $[\rho : z]$  are line coordinates and  $O = [0 : 1]$ . It contributes  $2n-4$  at  $O$ . Summing over the  $n$  lines gives  $2n(n-2)$ ; the remaining cycle is  $P \cdot S$ , proving (29). The total degree is  $3n(n-2)$ , as expected. By (12),

the central cluster is entirely nonreal for small real nonzero parameter, with multiplicities understood.

Formula (29) is an application of existing specialization theory. On the flat family  $X = \{P + tV = 0\}$ , put  $H = \det \text{Hess}(P + tV) = tK$ . The divisor identity is

$$\text{div}(H) = X_0 + \text{div}(K).$$

Since  $K_0$  contains no component of  $P = 0$ , Theorem 4.1 of [Esteves \(2010\)](#) applies with  $E_i = 0$ ,  $p_i = 1$  and  $F_i = \text{div}(K)$ . Equivalently, Proposition 5.2 of [Esteves et al. \(2021\)](#) applies with  $G = H$ ,  $L_i = 1$ ,  $M_i = K$ . The extra irreducibility assumption in ([Esteves, 2010](#), Example 5.3), and the nondegenerate-system assumptions of the specialized ramification formulas in ([Esteves et al., 2021](#)), do not obstruct these general applications. The complex cycle alone cannot distinguish real flexes approaching  $O$  from conjugate pairs. That distinction is supplied by the uniform real estimate in Section 2.

## AI assistance

OpenAI Codex contributed substantially to mathematical exploration and proof development, symbolic calculations and verification code, literature searches, and manuscript drafting and revision. The author directed the project and is responsible for the claims. The author reports using the most capable GPT model available to him in Codex at the time; exact historical model-version identifiers are not documented. The accompanying scripts reproduce the finite example checks. Those checks do not replace the general proofs, and AI review does not constitute independent specialist review or formal verification.

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